Roll No.: ……………..……………….

END SEMESTER EXAMINATION MAY 2025

Name of the Program: B.Tech | Semester: VIII

Name of the Course: Mobile Computing | Course Code: TOE 811

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Identify and explain the major challenges currently faced in the field of mobile computing. Provide a detailed discussion on each challenge, including its causes, and implications.	CO1
(b)	How does cell sectoring affect channel borrowing in cellular networks? Analyze its impact on co-channel interference and explain how it contributes to improved frequency reuse and overall network efficiency.	CO2
(c)	Compare and contrast mobile computing and distributed computing in terms of their architecture, characteristics, and application domains. Highlight the key differences in how they handle connectivity, resource sharing, and mobility.	CO2
Q2		
(a)	Define cellular network. In the context of cellular networks, explain the concepts of clustering, frequency reuse distance, cell splitting, and co-channel interference.	CO1
(b)	What is channel allocation in cellular systems? Elaborate on Dynamic Channel Allocation and distinguish between Fixed and Dynamic Channel Allocation strategies by highlighting their key differences.	CO1
(c)	In a cellular system, the total number of channels per cell is given as six and two channels are reserved exclusively for handoff calls. What are the blocking probabilities for originating if the handoff request rate is 0.0001, the originating call rate is 0.001, and the service rate μ=0.0003?	CO2
Q3		
(a)	Describe how IP packet delivery is handled in Mobile IP. Explain the processes of agent advertisement, agent discovery, and mobile node registration in detail.	CO2
(b)	Explain the architecture and design principles of the CODA file system. Discuss the various operational states of Venus in detail. Include a diagram to support your explanation.	CO3 CO3

(c)	What are the key data management issues in mobile computing? Discuss the major challenges associated with managing data in mobile environments.	
Q4		
(a)	What is data replication in the context of mobile computing? Explain the purpose, benefits, and challenges of implementing data replication in mobile environments. Discuss various replication strategies and their impact on system performance and data consistency.	CO3
(b)	Define mobile agent-based computing. Describe the life cycle of a mobile agent, and illustrate your explanation with an appropriate diagram.	CO4
(c)	Define mobile agents and explain how their unique properties differentiate them from conventional software programs. Also, discuss the key security requirements necessary for ensuring safe and reliable mobile agent systems.	CO4
Q5		
(a)	What are the essential characteristics of an ideal routing protocol for ad hoc wireless networks? Outline the key differences between proactive and reactive routing protocols. Additionally, explain the route discovery process in the Ad hoc On-Demand Distance Vector (AODV) routing protocol.	CO5&6 CO5&6
(b)	Describe the routing process in Mobile Ad Hoc Networks (MANETs). Additionally, provide a detailed explanation of the Destination Sequenced Distance Vector (DSDV) Routing protocol.	CO5&6
(c)	Compare and contrast the DSR (Dynamic Source Routing) and AODV (Ad hoc On-Demand Distance Vector) routing protocols with respect to routing overhead, the impact of mobility, routing types, and how they handle request replies.	